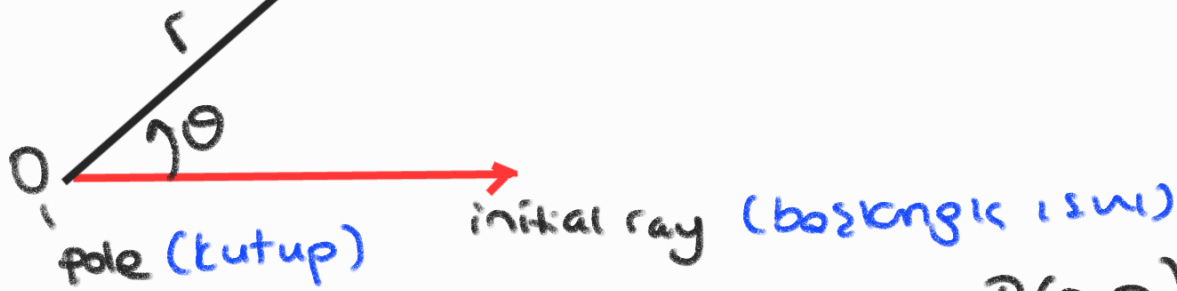


11.3 Polar Coordinates: (Kutupsal Koordinatlar)

$$P(x, y) = P(r, \theta)$$



r : positive & negative
 θ : positive & negative

$P(r, \theta)$
directed distance
from O to P

O ile P
arasındaki
yönlü
mesafe

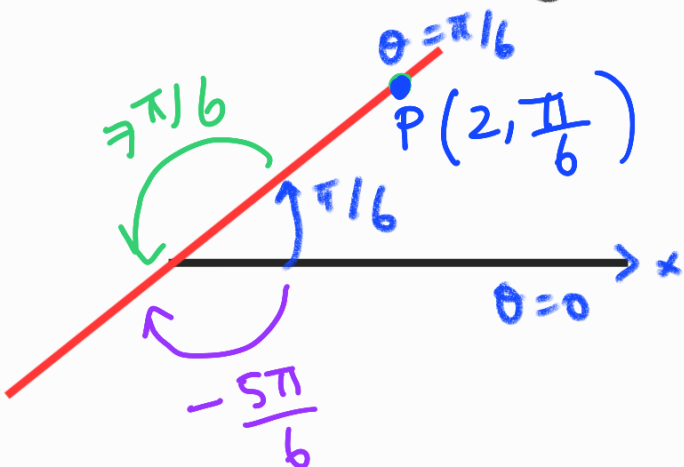
directed
angle
from
initial ray
to OP

Başlangıç
ışından
O'ya doğru
yönlü açı

$P(2, \frac{\pi}{6})$ noktasının tüm
kutupsal koordinatlarını
bulun.

Ex: Find all polar coordinates of the point $P(2, \frac{\pi}{6})$.

We sketch the initial ray of the coordinate system, draw the ray from the origin that makes an angle of $\frac{\pi}{6}$ with the initial ray and mark the point $(2, \frac{\pi}{6})$.



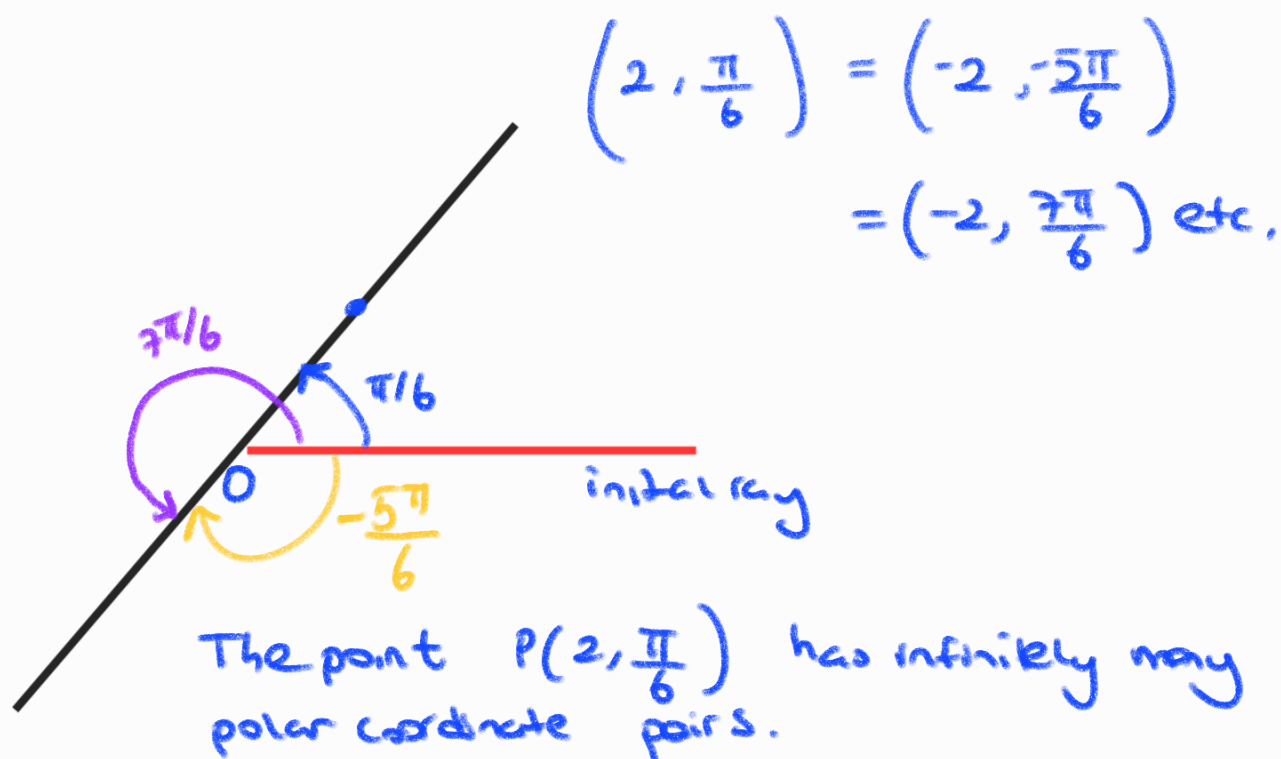
We find the angles for the
other coordinate pairs of P
in which $r=2$, and $r=-2$.

For $r=2$, the infinite list of angles is

$$\frac{\pi}{6}, \frac{\pi}{6} \pm 2\pi, \frac{\pi}{6}, \pm 4\pi, \frac{\pi}{6} \pm 6\pi$$

for $r=-2$, the angles are

$$-\frac{5\pi}{6}, -\frac{5\pi}{6} \pm 2\pi, -\frac{5\pi}{6} \pm 4\pi, -\frac{5\pi}{6} \pm 6\pi, \dots$$



The corresponding coordinate pair of P are

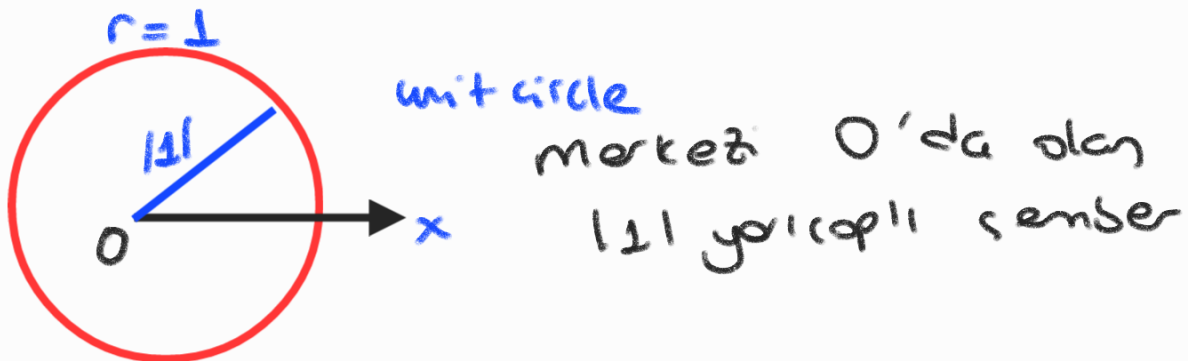
$$\left(2, \frac{\pi}{6} + 2n\pi\right), n=0, \pm 1, \pm 2, \dots$$

and

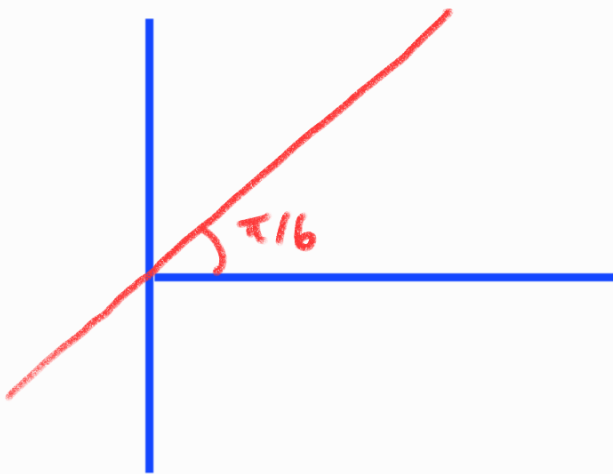
$$\left(-2, -\frac{5\pi}{6} + 2n\pi\right), n=0, \pm 1, \pm 2, \dots$$

Ex: (a) $r=1$ (θ changes) $\rightarrow \theta$ deđis:yor
 $r=-1$ (θ changes)

If we hold r fixed at a constant value $r=1$, the point $P(r, \theta)$ will lie $|r|=1$ units from the origin O . As θ varies over any interval of length 2π , P then traces a circle of radius $|r|=1$ centred at O .

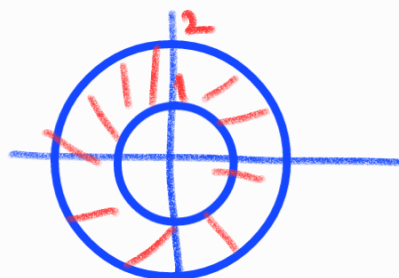


(b) $\theta = \frac{\pi}{6}$ (r changes) r deđis:ir
 Enter line
 doğru

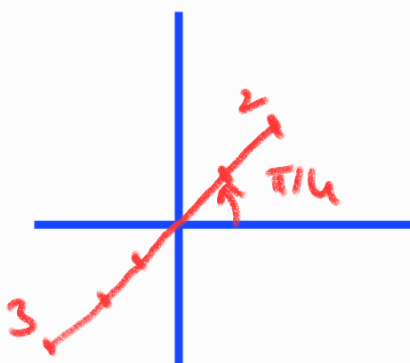


Ex: Graph the sets of points whose polar coordinates satisfy the following conditions.

(a) $1 \leq r \leq 2$



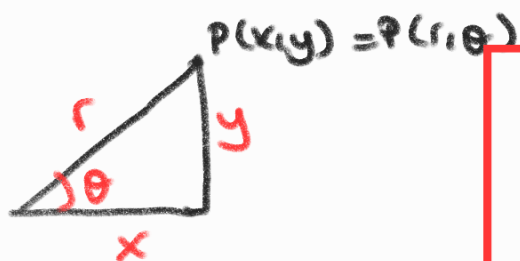
(b) $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$



Kutupool ve Kartezyen koordinatları ilişkilerinin Denklemleri:

Cartesian Versus Polar Coordinates:

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}$$



$$\begin{aligned} x &= r \cos \theta & x^2 + y^2 &= r^2 \\ y &= r \sin \theta & \frac{y}{x} &= \tan \theta \end{aligned}$$

Ex: Replace the following polar equations by equivalent Cartesian equations.

Polar denklemleri kartezyen denkleme çevirin.

(a) $r^2 \cos \theta \sin \theta = (r \cos \theta)(r \sin \theta) = xy$

$$r^2 \cos \theta \sin \theta = 4 \Rightarrow xy = 4$$

(b) $r = 1 + 2r \cos \theta \Rightarrow \sqrt{x^2 + y^2} = 1 + 2x$

$$\Rightarrow x^2 + y^2 = (1 + 2x)^2 \Rightarrow y^2 - 3x^2 - 4x - 1 = 0.$$

$$\textcircled{c} \quad r = 1 - \cos \theta$$

$$\Rightarrow \frac{r}{\sqrt{x^2 + y^2}} = 1 - \frac{x}{\sqrt{x^2 + y^2}}$$

$$\Rightarrow x^2 + y^2 = \sqrt{x^2 + y^2} - x$$

$$\Rightarrow (x^2 + y^2 + x)^2 = x^2 + y^2$$

$$\Rightarrow x^4 + y^4 + \underline{x^2} + 2x^2y^2 + 2x^3 + 2xy^2 = \underline{x^2 + y^2}$$

Ex: Find a polar equation for the circle

$$x^2 + (y-3)^2 = 9$$

centerin kutupda degeni
söylün.

$$r = 3, \text{ center : } (0, 3)$$

$$(x-x_0)^2 + (y-y_0)^2 = r^2$$

$$\Rightarrow x^2 + y^2 - 6y + 9 = 9$$

$$\Rightarrow x^2 + y^2 - 6y = 0 \Rightarrow r^2 = 6r \sin \theta$$

$$\Rightarrow r = 6 \sin \theta$$

Ex! (a) Convert $(\underbrace{-4}_r, \underbrace{\frac{2\pi}{3}}_\theta)$ into Cartesian coordinates.

Verilen polar koordinatları Kartezyene çevirin.

$$x = r \cos \theta = -4 \cos \frac{2\pi}{3} = -4 \left(-\frac{1}{2}\right) = 2$$

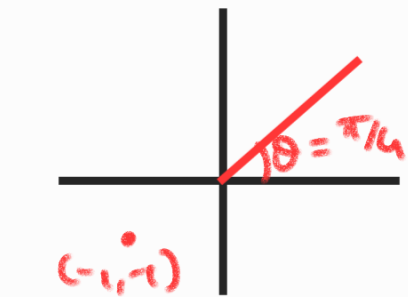
$$y = r \sin \theta = -4 \sin \frac{2\pi}{3} = (-4) \frac{\sqrt{3}}{2} = -2\sqrt{3}$$

$$\Rightarrow (x, y) = (2, -2\sqrt{3})$$

⑥ Convert $(-1, -1)$ into polar coordinates. Polar coordinates rev'n.

$$r = \sqrt{x^2 + y^2} = \sqrt{2}, \quad \theta = \tan^{-1} \frac{y}{x}$$

$$= \tan^{-1}(1) = \frac{\pi}{4},$$



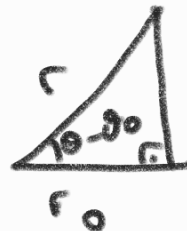
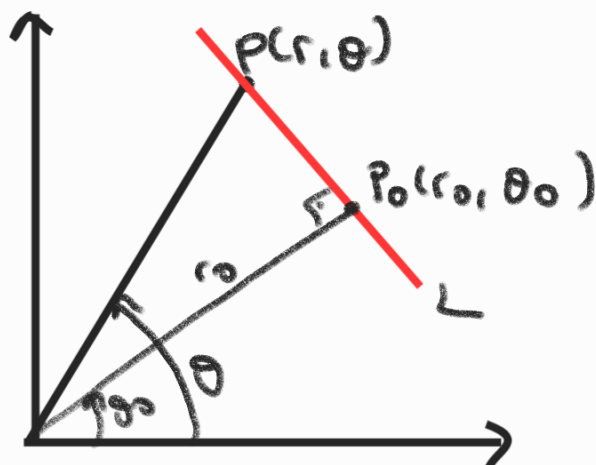
$(-1, -1)$
is in 3rd quadrant.

$$\theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

$$\Rightarrow (\sqrt{2}, \frac{5\pi}{4})$$

Kutupsal Koordinatlerde Doğrular ve Çemberler:

Lines and Circles:



$$\cos(\theta - \theta_0) = \frac{r_0}{r}$$

$$r \cos(\theta - \theta_0) = r_0$$

(r, θ) unknowns!

Örjinder L doğrudan
gelen dikmenin ortağı!

$P_0(r_0, \theta_0)$ noktası ise
ve $r_0 \geq 0$ ise, L doğrusu
için denklem:

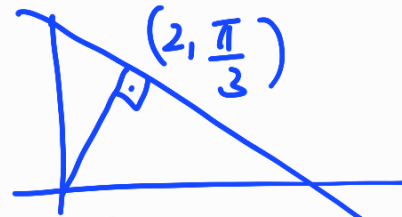
Ex: $P_0(2, \frac{\pi}{3})$ given.

$$r \cos(\theta - \theta_0) = r_0$$

$$r \cos(\theta - \frac{\pi}{3}) = 2$$

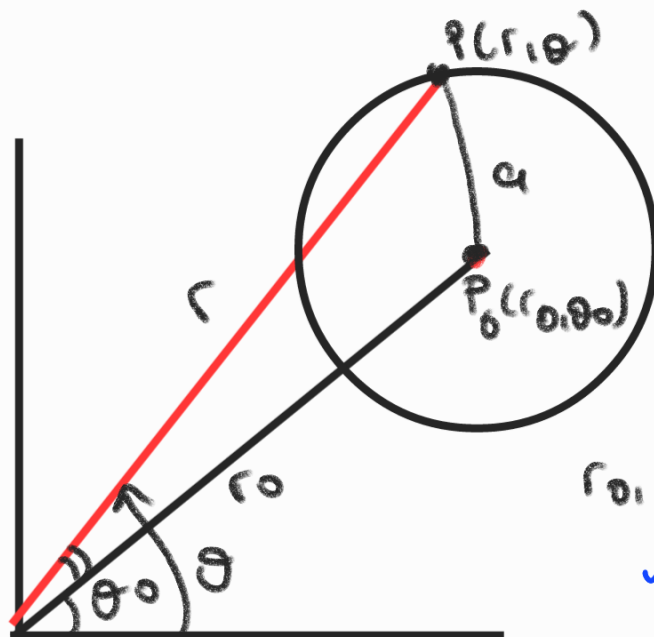
$$r \left(\cos\theta \cdot \cos\frac{\pi}{3} + \sin\theta \sin\frac{\pi}{3} \right) = 2 \quad L$$

$$\frac{1}{2} r \cos\theta + \frac{\sqrt{3}}{2} r \sin\theta = 2$$



$$\frac{1}{2} x + \frac{\sqrt{3}}{2} y = 2 \Rightarrow x + \sqrt{3} y = 4$$

Circles:
Çemberler



r_0, θ_0 given.
veriliyor

$$a^2 = r_0^2 + r^2 - 2r_0 r \cos(\theta - \theta_0)$$

Law of cosines

Kosinüs kuralından ($O P_0 P$ üçgeni için)

Center merkezinde, gerçekte, $r_0 = a$ olur.
 If the circle passes through origin, then
 $r_0 = a$

$$a^2 = a^2 + r^2 - 2ar \cos(\theta - \theta_0)$$

$$r = 2a \cos(\theta - \theta_0)$$

If the circle's center lies on the positive
 x-axis $\theta_0 = 0$

$$\Rightarrow r = 2a \cos \theta$$

Centerin merkezi
 poz. x-ekseni
 üzerinde olduğu, $\theta_0 = 0$
 olur.

If the circle's center lies on the positive y-axis
 $\theta_0 = \frac{\pi}{2}$

Centerin merkezi poz.
 y-ekseni etrafında

$$\cos\left(\theta - \frac{\pi}{2}\right) = \sin \theta \quad \text{olursa, } \theta_0 = \frac{\pi}{2} \text{ olur.}$$

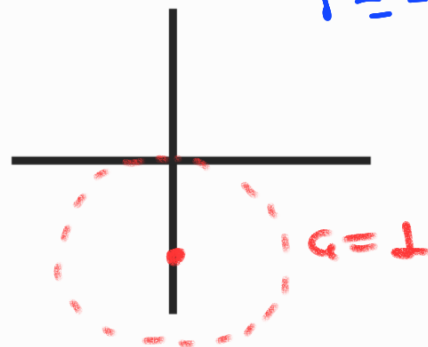
$$\Rightarrow r = 2a \sin \theta$$

* merkez neg.
 x-ekseninde: $r = -2a \cos \theta$
 merkez neg. y-ekseni:
 $r = -2a \sin \theta$

Ex: $\left(-1, \frac{\pi}{2}\right)$

Kutup
 koordinatları
 centerin
 denklemini

center in
 polar coord.



$$r = -2 \sin \theta$$

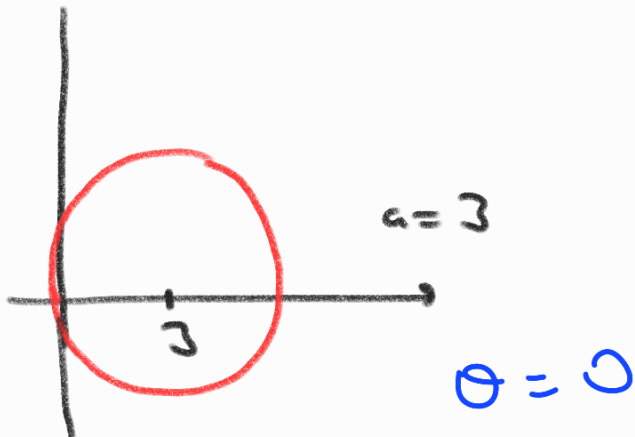
Equation in polar
 coordinates

Ex:

1) $r = 6 \cos \theta$

yarıçap: 3, merkez $(3, 0)$

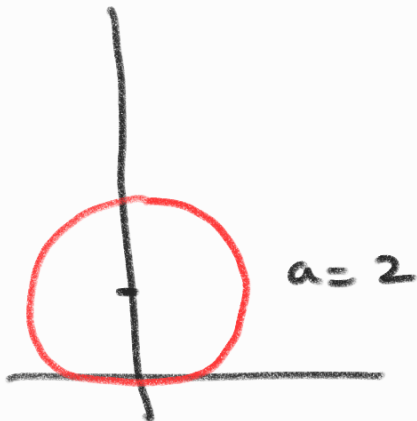
$\Rightarrow$ radius = 3, center in polar coordinates $(3, 0)$



2) $r = 4 \sin \theta$

yarıçap: 2, merkez: $(2, \frac{\pi}{2})$

$\Rightarrow$ radius = 2, center in polar coordinates $(2, \frac{\pi}{2})$



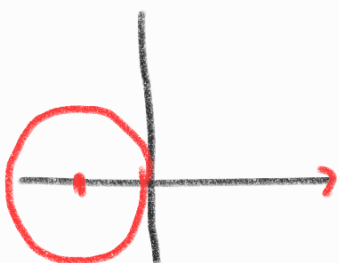
3) $r = -\cos \theta$

yarıçap

$\Rightarrow a = \frac{1}{2}$ (radius)

merkez: $(-\frac{1}{2}, 0)$
veya

center in polar coordinates
 $(\frac{1}{2}, \pi)$
: $(-\frac{1}{2}, 0)$ kut.
koordinatları



Kutupsal Koordinatlar Grafik Çizmek

11.4 Graphing in Polar Coordinates

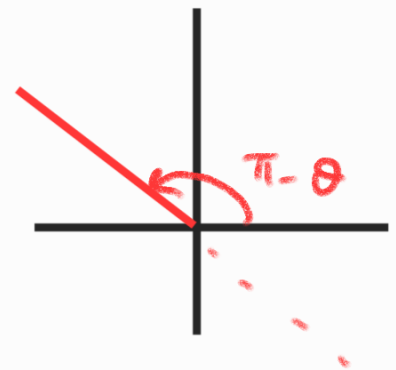
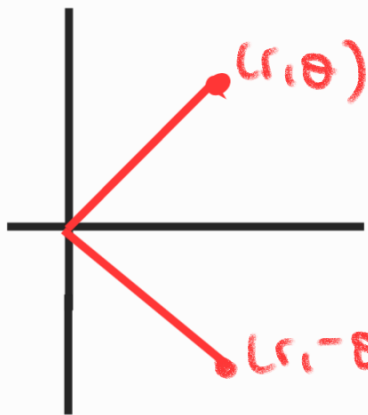
Symetry Simetri:

x-eksenine göre simetri:

1) Sym. about the x-axis: If the point (r, θ)

lies on the graph, the point $(r, -\theta)$ or

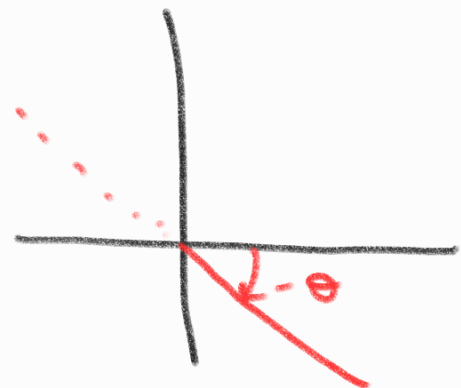
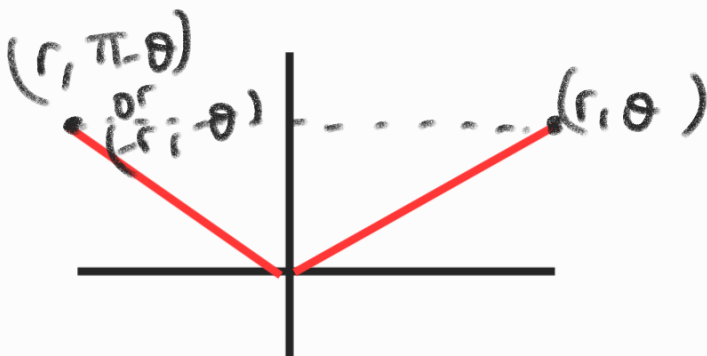
$(-r, \pi - \theta)$ lies on the graph.



2) Symetry about the y-axis: If the point

(r, θ) lies on the graph the point $(r, \pi - \theta)$

or $(-r, -\theta)$ lies on the graph.



Simetri:

x-eksenine göre simetri: (r, θ) noktası

grafik üzerindeyse, $(r, -\theta)$ veya $(-r, \pi - \theta)$ noktası da grafik üzerindedir.

y-eksenine göre simetri: (r, θ) noktası

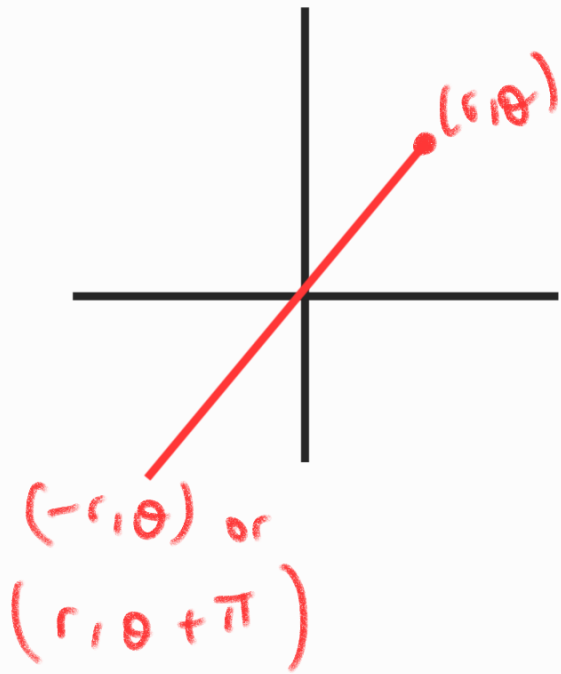
grafik üzerindeyse, $(r, \pi - \theta)$ veya $(-r, \theta)$ noktası da grafik üzerindedir.

oriijine göre simetri: (r, θ) noktası grafik

üzerindeyse, $(-r, \theta)$ veya $(r, \theta + \pi)$ noktası da grafik üzerindedir.

3) Symetric about the origin: If the point

(r, θ) lies on the graph, the point $(-r, \theta)$ or $(r, \theta + \pi)$ lies on the graph.



$$\frac{dy}{d\theta} = \frac{dy}{dx} \frac{dx}{d\theta}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

Slope of a Polar Curve ^{egim} $r = f(\theta)$

$$x = r \cos \theta = f(\theta) \cos \theta$$

$$y = r \sin \theta = f(\theta) \sin \theta$$

$$m = \frac{dy}{dx} \bigg|_{(r, \theta)}$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{df}{d\theta} \sin \theta + f(\theta) \cos \theta}{\frac{df}{d\theta} \cos \theta - f(\theta) \sin \theta}$$

$\left(\frac{dx}{d\theta} \neq 0 \text{ olmak lazımlıdır} \right)$

Ex: Write the eqn. of the tangent line to

$$r = 3 + 8 \sin \theta \quad \text{at} \quad \theta = \frac{\pi}{6}.$$

$$\frac{dy}{dx} = \frac{8 \cos \theta \cdot \sin \theta + (3 + 8 \sin \theta) \cos \theta}{8 \cos^2 \theta - (3 + 8 \sin \theta) \sin \theta}$$

$$m = \left. \frac{dy}{dx} \right|_{\theta = \frac{\pi}{6}} = \frac{4\sqrt{3} + \frac{3\sqrt{3}}{2}}{4 - \frac{3}{2}} = \frac{11\sqrt{3}}{5}$$

$$\theta_0 = \frac{\pi}{6} \Rightarrow r_0 = 7 \quad (r_0 = 3 + 8 \sin \theta_0)$$

$$x_0 = 7 \cos \frac{\pi}{6} = 7 \frac{\sqrt{3}}{2}, \quad y_0 = 7 \sin \frac{\pi}{6} = \frac{7}{2}$$

$$x_0 = f(\theta_0) \cos \theta_0, \quad y_0 = f(\theta_0) \sin \theta_0$$

$$y = \frac{7}{2} + \frac{11\sqrt{3}}{5} \left(x - \frac{7\sqrt{3}}{2} \right)$$

Ex: $r = 1 - \cos \theta$ (cardioid)

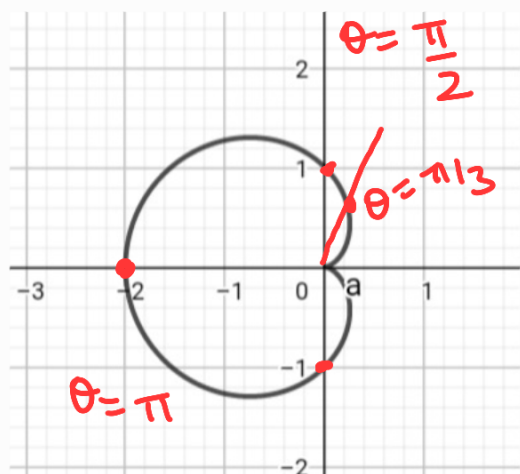
$$(r, \theta) \text{ on the graph} \Rightarrow r = 1 - \cos \theta$$

$$r = 1 - \cos(-\theta)$$

$$\Rightarrow (r, -\theta) \text{ on the graph}$$

$$[\cos \theta = \cos(-\theta)]$$

$\Rightarrow$ sym. about the x-axis.



$$\theta = 0$$

θ	r
0	0
$\pi/3$	$1/2$
$\pi/2$	1
π	2

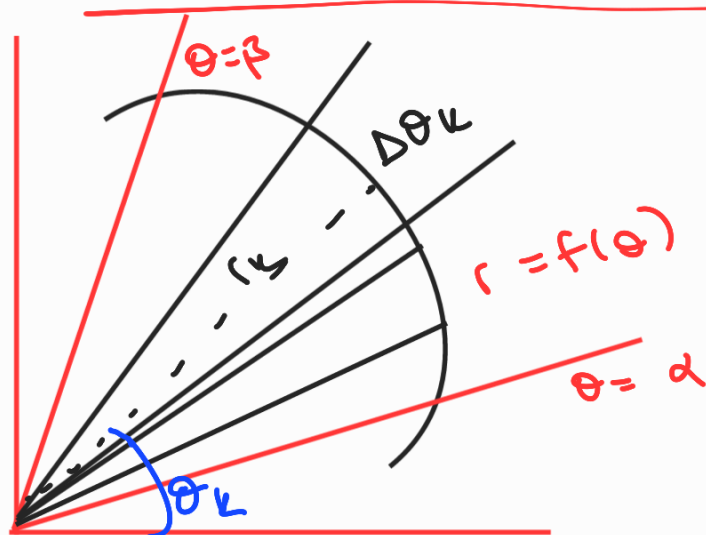
Ex: Find the slope of the cardioid

$$r = -1 + \cos \theta \quad \text{at} \quad \theta = \frac{\pi}{2}.$$

$$m = \left. \frac{dy}{dx} \right|_{\theta = \frac{\pi}{2}} = \frac{-1 \cdot \sin \frac{\pi}{2} + (-1) \cos \frac{\pi}{2}}{-1 \cdot \cos \frac{\pi}{2} - (-1) \sin \frac{\pi}{2}} = -1$$

$$\begin{aligned} f(\theta) &= -1 + \cos \theta & , & \quad f\left(\frac{\pi}{2}\right) = -1 \\ f'(\theta) &= -\sin \theta & , & \quad f'\left(\frac{\pi}{2}\right) = -1. \end{aligned}$$

Area of the Polar Curve:



$$A_k = \frac{1}{2} r_k^2 \Delta \theta_k$$

$$\left(A = \frac{1}{2} r^2 2\pi = r^2 \pi \right)$$

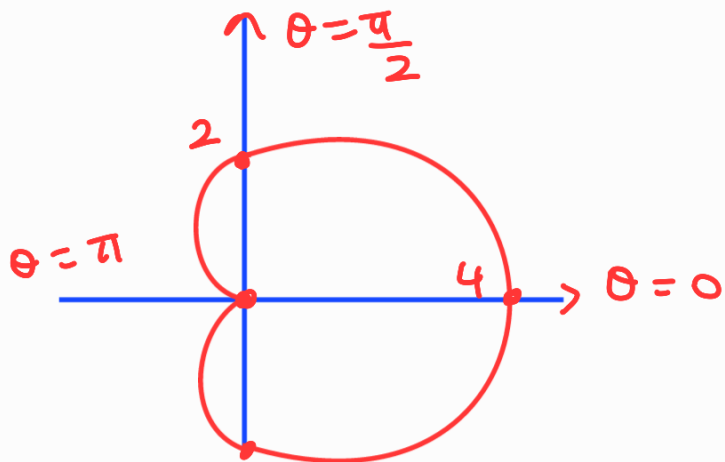
$$\lim_{n \rightarrow \infty} \sum_{k=1}^n A_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{2} f(\theta_k)^2 \Delta \theta_k$$

$$\Rightarrow A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

Ex: Find the area of the region

enclosed by the cardioid $r = 2(1 + \cos \theta)$

θ	r
0	4
$\pi/2$	2
π	0



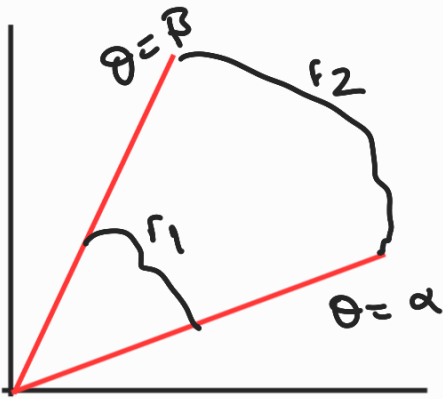
$$A = \int_0^{2\pi} \frac{1}{2} 4 (1 + \cos \theta)^2 d\theta$$

$$= \int_0^{2\pi} 2 (1 + 2\cos \theta + \cos^2 \theta) d\theta$$

$$\begin{aligned}
&= \int_0^{2\pi} \left(2 + 4 \cos \theta + 2 \frac{1 + \cos 2\theta}{2} \right) d\theta \\
&= \int_0^{2\pi} (3 + 4 \cos \theta + \cos 2\theta) d\theta \\
&= 3\theta + 4 \sin \theta + \frac{\sin 2\theta}{2} \bigg|_0^{2\pi} = 6\pi.
\end{aligned}$$

Area of the Region

$$\begin{aligned}
0 &\leq r_1(\theta) \leq r \leq r_2(\theta), \\
\alpha &\leq \theta \leq \beta
\end{aligned}$$



$$\begin{aligned}
A &= \int_{\alpha}^{\beta} \frac{1}{2} r_2^2 d\theta - \int_{\alpha}^{\beta} \frac{1}{2} r_1^2 d\theta \\
&= \int_{\alpha}^{\beta} \frac{1}{2} (r_2^2 - r_1^2) d\theta
\end{aligned}$$

r_2 : outer curve

r_1 : inner curve